THE UNIVERSITY OF LAHORE

Department of Computer Science & IT
Faculty of Information Technology

Final Year Project Proposal

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PROJECT TITLE:	Integrated Health Information System (IHIS)
KEY WORDS:	PRMS, EHR, Healthcare IT, Public Healthcare
DOMAIN OF THE PROJECT:	Electronic Health Record
SUPERVISOR'S NAME:	Sir Asif Ahsan

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PROBLEM STATEMENT

The current healthcare system in Pakistan suffers a lot of inefficiency due to lack of interoperability. Paper-based records, siloed within individual healthcare institutions, hinder seamless patient care, impede research, and limit the government's ability to effectively plan and deliver public health services

EXECUTIVE SUMMARY

This project aims to develop a secure and interoperable Web application and Mobile app(for patient portal) for Centralized Patient Record Management System (C-PRMS) for government healthcare institutions in Pakistan. Recognizing the limitation of fragmented patient data, the C-PRMS is expected to improve healthcare delivery by empowering healthcare professionals and patients.
Key user stories include:

"As a doctor, I would like to have access to a patient's entire medical history from all government hospitals so that I can make the right diagnoses and treatment decisions."
"As a nurse, I would like to be able to update a patient's medical records electronically so that I can ensure that all healthcare providers have the right information at the right time."
"As a public health official, I want to analyse aggregated patient data in order to identify disease outbreaks and track public health trends so that I can effectively plan and implement public health interventions."
"As a patient, I want to securely access and view my own medical records online so that I can be an active participant in my healthcare decisions and share information with other healthcare providers."

Using the latest technologies, such as cloud computing, micro-services architecture, and blockchain, the C-PRMS will ensure the data's security, privacy, and interoperability and deliver a user-centric impactful solution for the Pakistani healthcare system.

INTRODUCTION

The healthcare system of Pakistan is still suffering from serious issues, and one of the most significant is fragmented patient data. Paper-based records, confined to individual health care institutions, create barriers for seamless patient care, hinder research, and the government's capacity to plan and deliver public health services.

Relevance:

- Fragmented data leads to delayed diagnoses, medication errors, and inconsistent treatment plans, thereby affecting the outcome for patients.

- Lack of interoperability is one of the challenges facing public health initiatives, where tracking disease outbreaks, planning resource allocation, and effective public health surveillance are hard to achieve.
- Poor data management burdens the administrative task of healthcare providers, diverting valuable time and resources from patient care.

Background:

- The Government of Pakistan has been actively promoting digital health initiatives to modernize healthcare delivery.
- The National Health Vision 2016-2025 focuses on the need to improve data quality and leverage technology to enhance healthcare services.
- However, the adoption of EHRs has been patchy, and a centralized system for data sharing across institutions is still largely absent.

Related Work:

- **Global Examples:**

  i. The UK's National Health Service (NHS) has successfully implemented a national electronic health record system, improving patient care and public health outcomes.

  ii. The United States has seen advancements in EHR systems, although interoperability challenges persist.

COMPETITORS/COMPETITIVE ANALYSIS

Potential Internal Competition:

HIS Implemented by Government Hospitals: Most government hospitals might have implemented some kind of very basic HIS. Their HISs could be very narrow in functionality, not interoperable, and would create challenges in the integration of a centralized system.

Data Silos: The presence of isolated data systems within separate hospitals creates major barriers to sharing and integrating data.

OBJECTIVES

This project aims to achieve the following key objectives:

1. **Develop a secure and interoperable Centralized Patient Record Management System (C-PRMS) for government healthcare institutions in Pakistan.** This system will facilitate the seamless exchange of patient data across all levels of the healthcare system.
2. **Enhance patient care quality and safety.** By providing healthcare providers with access to complete and accurate patient information, the C-PRMS will enable more informed clinical decision-making, reduce medical errors, and improve patient outcomes.
3. **Improve public health outcomes.** The C-PRMS will support public health initiatives by enabling:
    - **Disease surveillance and outbreak response:** Timely identification and tracking of infectious diseases.
    - **Resource allocation:** Informed planning and allocation of healthcare resources based on population needs.
    - **Public health research:** Facilitate research and analysis to improve public health policies and interventions.
4. **Increase efficiency and reduce administrative burden.** The C-PRMS will streamline administrative processes, reduce paperwork, and improve the efficiency of healthcare delivery within government institutions.
5. **Empower patients:** Provide patients with secure access to their health records, enabling them to actively participate in their healthcare decisions and improve their health literacy.
6. **Strengthen the public health infrastructure:** Establish a robust foundation for a more data-driven and efficient healthcare system in Pakistan.

MOTIVATION

This project presents a unique and challenging opportunity for software engineers to contribute to a critical national initiative. The development of a robust and interoperable Centralized Patient Record Management System (C-PRMS) for government healthcare institutions in Pakistan demands a high level of technical expertise and innovative problem-solving.

From a software engineering perspective, this project offers:

- **An opportunity to work on a large-scale, complex system:** The C-PRMS will involve the design and development of a sophisticated system with high availability, scalability, and security requirements.
- **The chance to apply cutting-edge technologies:** This project will require the implementation of modern technologies such as cloud computing, microservices architecture, blockchain, and artificial intelligence, providing valuable hands-on experience with these emerging technologies.
- **The opportunity to solve challenging technical problems:** Addressing the challenges of data security, privacy, interoperability, and scalability will require creative problem-solving and innovative technical solutions.

Furthermore, successful completion of this project will have a significant and lasting impact on the Pakistani healthcare system. By contributing to the development of this critical national infrastructure, software engineers can directly contribute to improving the quality of life for millions of Pakistani citizens.

This project provides a unique and rewarding opportunity for software engineers to apply their skills and expertise to a challenging and impactful project with the potential to make a real difference in the world.

FEATURES OF PROJECT

1. Patient Registration and Demographics:

- **Unique Patient Identifier:** Assigning a unique and persistent identifier to each patient across all government healthcare institutions, enabling accurate patient identification and preventing duplicate records. This will be achieved through integration with the NADRA database.
- **Comprehensive Demographic Data:** Capturing essential demographic information, including full name, date of birth, gender, address, contact information, emergency contacts, and other relevant socio-demographic data.

2. Electronic Health Records (EHR) Management:

- **Secure Storage and Retrieval:** Securely storing and retrieving complete patient medical histories, including past medical conditions, surgeries, medications, allergies, immunizations, laboratory results, and radiology images.
- **Clinical Documentation:** Enabling electronic documentation of patient encounters, including consultations, diagnoses, treatment plans, procedures, and medications. This will include support for structured data entry, clinical decision support systems (e.g., drug interaction warnings), and electronic prescriptions.
- **Medication Management:** Facilitating electronic prescribing, medication reconciliation, and tracking of medication adherence to minimize medication errors and ensure patient safety.

3. Appointment Scheduling and Management:

- **Online Appointment Scheduling:** Enabling patients to book appointments online, reducing wait times and improving access to care.
- **Appointment Reminders:** Automated appointment reminders via SMS or email to improve patient adherence and reduce no-show rates.
- **Waiting List Management:** Efficiently managing waiting lists and minimizing patient waiting times.

4. Public Health Surveillance:

- **Disease Reporting:** Enabling real-time reporting of notifiable diseases and other public health concerns.

- **Outbreak Detection and Response:** Facilitating early detection and rapid response to disease outbreaks through data analysis, real-time alerts, and automated reporting mechanisms.
- **Surveillance Data Analysis:** Generating reports and dashboards to track public health trends, identify areas for improvement, and inform public health interventions.

5. User Management and Access Control:

- **Role-Based Access Control (RBAC):** Implementing a robust RBAC system to grant appropriate access levels to different user roles (doctors, nurses, administrators, public health officials, etc.).
- **Audit Trails:** Maintaining detailed audit trails of all user activities within the system to ensure data integrity, accountability, and compliance with data privacy regulations.

6. Data Security and Privacy:

- **Robust Security Measures:** Implementing robust security measures, including encryption, access controls, intrusion detection systems, and regular security audits, to protect sensitive patient data.
- **Data Privacy Compliance:** Ensuring strict adherence to all relevant data privacy laws and regulations, including the Data Protection Act 2020.

7. System Administration and Maintenance:

- **System Monitoring and Maintenance:** Implementing robust system monitoring and maintenance procedures, including regular backups, software updates, and performance tuning.
- **Disaster Recovery and Business Continuity Planning:** Developing and implementing a comprehensive disaster recovery and business continuity plan to ensure system availability and data integrity in the event of emergencies.

8. Integration with External Systems(Optional):

- **NADRA Integration:** Seamless integration with NADRA databases for accurate and reliable patient identification and address verification.
- **Interoperability with Other Systems:** Enabling seamless data exchange with other government systems, such as the national immunization registry, the national drug regulatory authority, and other relevant health information systems.

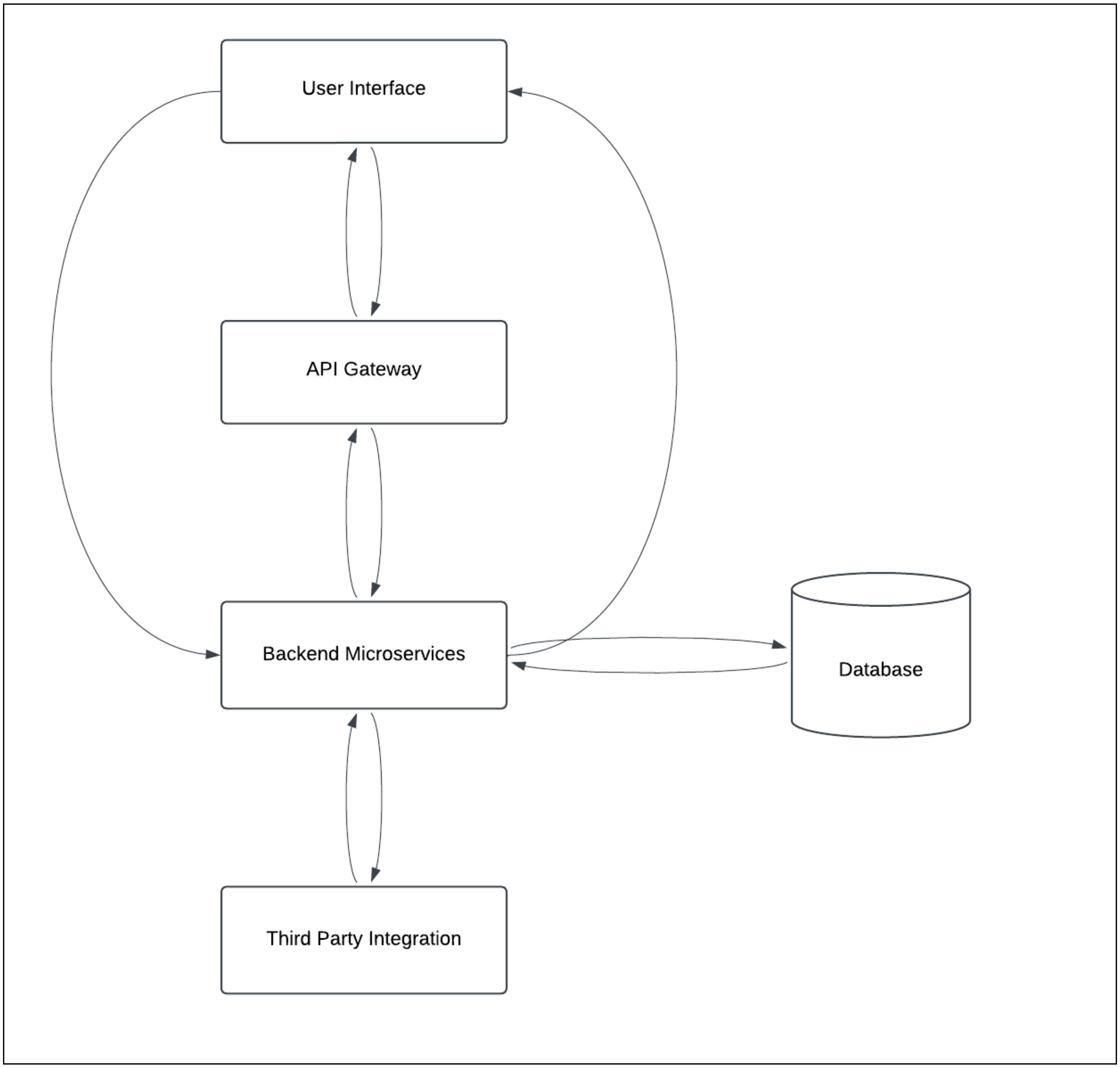
9. Patient Portal:

- **Secure Patient Portal:** Providing patients with secure access to their health records online, enabling them to view appointment schedules, request prescription refills, communicate with their healthcare providers, and access health education materials.

10. Blood Bank:

- **Blood Bank Feature** : The blood bank feature is a comprehensive module designed to streamline the management of blood donation and distribution processes. It enables users to search for specific blood groups, check availability, and locate nearby blood banks with real-time updates on stock levels. Donors can register their details, schedule donations, and track their donation history, while recipients can request blood or connect with potential donors during emergencies. The system also ensures secure storage of donor and recipient information and generates reports for efficient inventory monitoring. By integrating this feature, the platform aims to improve accessibility to life-saving blood resources and enhance communication between blood banks, donors, and recipients.

ARCHITECTURAL DESIGN



IMPLEMENTATION TOOLS AND TECHNIQUES

Methodology:

The C-PRMS will be developed using an Agile methodology, specifically Scrum, to ensure flexibility, iterative development, and continuous improvement.

- **Scrum Framework:**
  - The project will be divided into short sprints (e.g., 2-week sprints) with clearly defined goals and deliverables.

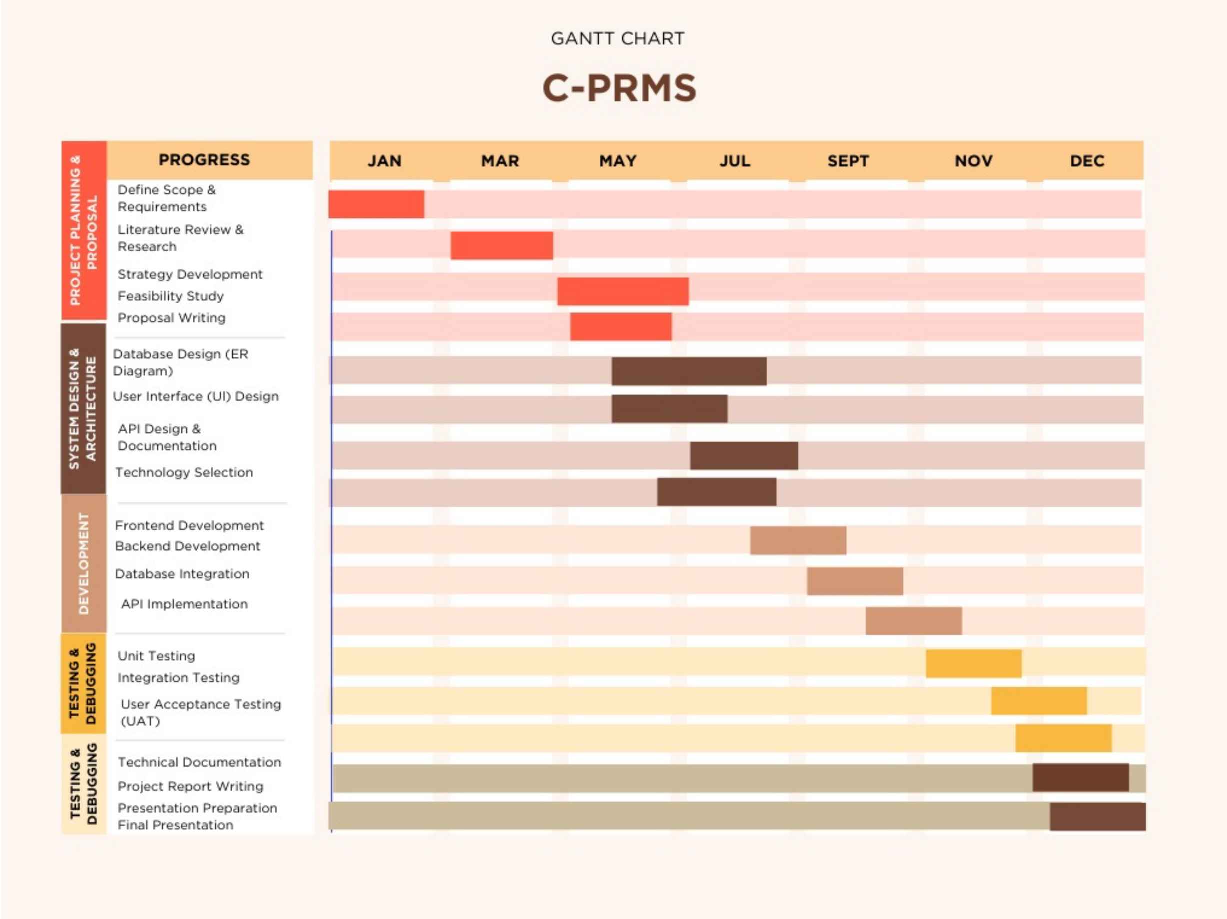
    - Sprint reviews and retrospectives will be conducted to assess progress, identify areas for improvement, and adjust the project plan as needed.
- **User-Centered Design:**
    - User stories will be created and managed using tools like **Jira** or **Azure DevOps**.
    - User feedback will be actively sought and incorporated throughout the development process to ensure that the C-PRMS meets the specific needs and expectations of its users.
- **Iterative Development:**
    - The system will be developed iteratively, with frequent releases of new features and functionalities.

Tools and Technologies:

- **Project Management:** Jira, Azure DevOps, Trello
- **Version Control:** Git (hosted on GitHub)
- **Cloud Platform(Optional):** AWS, Azure, or GCP
- **Containerization(Optional):** Docker and Kubernetes
- **Programming Languages:** JavaScript/TypeScript.
- **Frontend Framework:** React and React-Native.
- **Backend Framework:** Node.js (with Express.js)
- **Database:** PostgreSQL (or a cloud-based database service)
- **Testing Frameworks:** JUnit, pytest, Mocha (for unit testing)
- **CI/CD Tools:** GitHub Actions for continuous integration and continuous delivery

Further tools will be added for security and data handling such as jwt, form hook,etc

PROJECT PLAN



REFERENCES

• World Health Organization (WHO) – Health Information Systems

- WHO emphasizes the importance of integrated health information systems in improving healthcare efficiency and decision making.

• National Institutes of Health (NIH) – Electronic Health Records (EHR) and Interoperability

- Discusses the benefits of electronic health records and data-sharing across healthcare facilities.

• HealthIT.gov – The Importance of Interoperability in Healthcare

- Explains how integrated health information systems enhance patient care and reduce medical errors

Supervisor's Signature: ………………………

FYP Proposal Evaluation

(To be filled by students)

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FOR EVALUATOR'S ONLY	
Remarks:	☐ Accepted ☐ Accepted with Minor changes ☐ Rejected

Suggested Improvements (*if any*):

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Reviewed By:

Name: ____

Signature: ____

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